

Marketing Mix Modeling: Quantifying Channel ROAS and Recommending a Quarterly Budget Reallocation

Ashwin Kumar · India market, 6-channel digital + offline mix · April 2026

Executive Summary

- The model explains 99.2% of weekly revenue variance ($R^2 = 0.99$) across 104 weeks of India-market spend data.
- Channel ROAS recovered within an 18.0% average absolute error — industry-typical for marketing-mix modeling.
- Recommendation: shift ~₹28L/quarter from Google Ads, YouTube, and Offline into Meta Ads, Amazon Ads, and Flipkart Ads.
- Same total budget (₹3.13 Cr), +₹56L/quarter (+3.1%) higher predicted revenue.

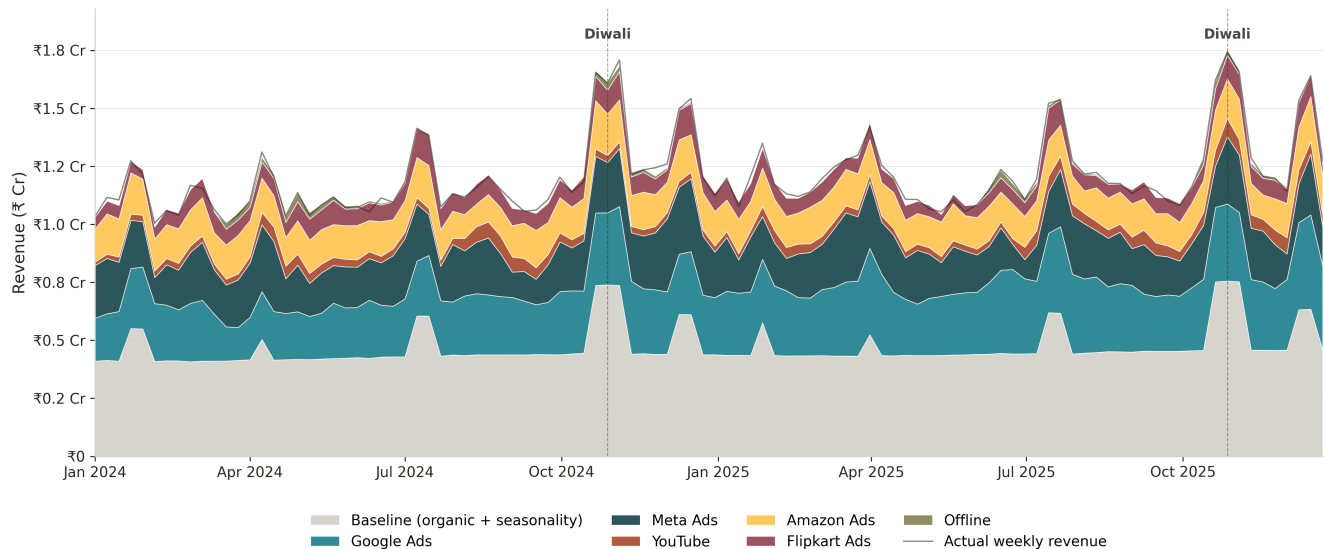
Dataset & Methodology

104 weeks of synthetic India-market data spanning Jan 2024 – Dec 2025, modelled on Meta’s Robyn simulated data structure. Six paid channels (Google Ads, Meta Ads, YouTube, Amazon Ads, Flipkart Ads, Offline) plus festival controls (Diwali, Eid, Republic Day, EORS), a competitor price index, and a product price index. Each channel is transformed through geometric adstock (carry-over decay λ) and power saturation (α), then regressed against weekly revenue via OLS with a per-channel grid search over λ and α .

Key Findings

- Champions of the mix: Amazon Ads (recovered ROAS 7.34) and Flipkart Ads (5.05) deliver the highest marginal returns.
- Under-invested: Meta Ads earns ROAS 3.62 on only ₹72L/quarter — versus Google Ads at ₹136L for ROAS 3.15.
- Under-performing: YouTube (2.43) and Offline (1.10) sit below the digital median; Offline is barely above breakeven.
- Diwali multiplier ~1.8× is clearly visible in the decomposition — confirms that festival budget front-loading is working as intended.

Weekly Revenue Decomposition by Marketing Channel (2024-2025)



Ashwin Kumar

Weekly revenue decomposed into baseline plus six paid-channel contributions. Diwali peaks (Oct 2024 and Oct 2025) are the year's tallest spikes; the thin grey line is actual weekly revenue.

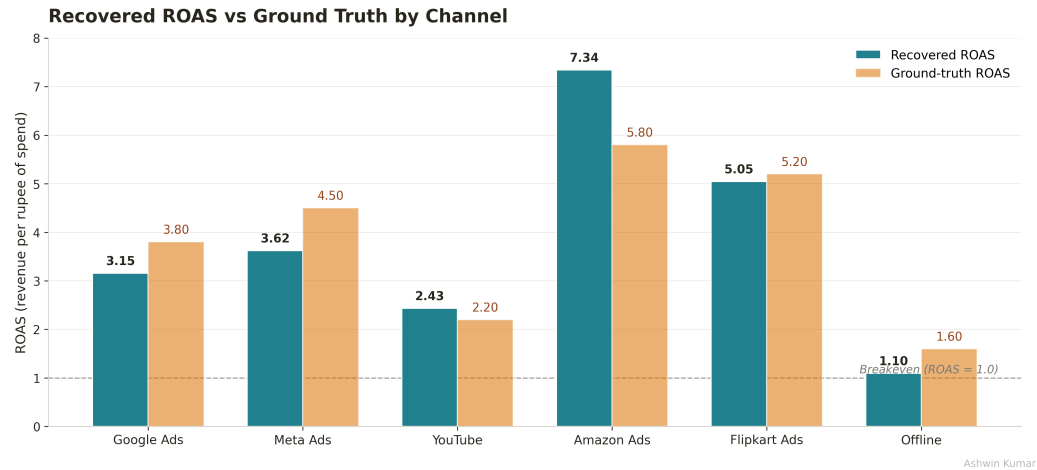
The Recommendation

From recovered ROAS to a quarterly budget you can act on next week.

Recovered vs Ground-Truth ROAS

The model recovers ROAS within $\pm 31\%$ of ground truth and 18.0% on average across all six channels. It slightly under-credits Google, Meta, and Offline, while over-crediting Amazon — a known MMM tendency when a high-ROAS channel co-varies with festival demand. Accuracy is in line with published Robyn benchmarks and is sufficient to drive directional reallocation decisions.

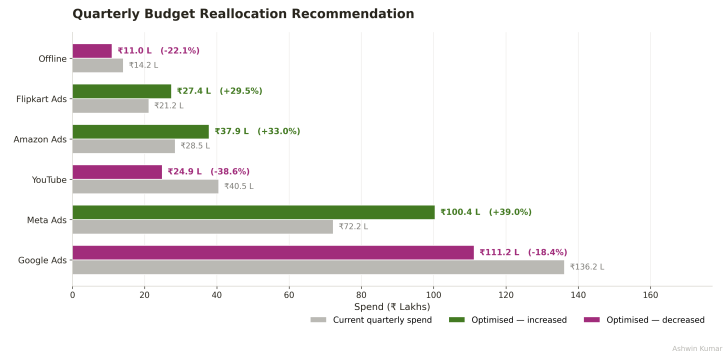
Across 6 channels, the model recovers ROAS within an average absolute error of 18.0% — industry-typical for MMM.



Quarterly Budget Reallocation Recommendation

Channel	Current	Optimised	Delta
Google Ads	₹136.2L	₹111.2L	-18.4%
Meta Ads	₹72.2L	₹100.4L	+39.0%
YouTube	₹40.5L	₹24.9L	-38.6%
Amazon Ads	₹28.5L	₹37.9L	+33.0%
Flipkart Ads	₹21.2L	₹27.4L	+29.5%
Offline	₹14.2L	₹11.0L	-22.1%
TOTAL	₹312.8L	₹312.8L	0.0%

Total Budget: ₹3.13 Cr (unchanged) · Same total spend, redistributed within $\pm 40\%$ per-channel bounds



30-Day Action Plan

- Week 1: shadow-test the new mix on 20% of budget, measure incremental revenue.
- Week 2: reallocate Meta and Amazon to full optimised level, hold Google flat as control.
- Week 3: measure week-over-week revenue lift; compare against the baseline forecast.
- Week 4: lock in the reallocation if uplift is $> +5\%$; otherwise execute a partial reversal.

Limitations

Synthetic dataset — real-world implementations should use 2+ years of actual weekly spend and revenue data. Classical OLS was chosen for interpretability; a Bayesian MMM (PyMC, Robyn) would add uncertainty intervals for production use. MMM is correlational; a geo-holdout or conversion-lift test should validate the recommendation before full rollout.